

Liu Yuchen (*Lyrice*)

Founder, Φ (fight) Research · Independent Researcher

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RESEARCH

Four papers submitted to NeurIPS 2026, anonymous code released (3 sole-author, 1 co-first).

Phantom Phases: How Attention Sinks Generate Fictitious Training Dynamics

Sole author · [desink-57F5](#)

Standard spectral metrics on transformer hidden states (E_1 , effective rank, RankMe, CKA, anisotropy) are systematically contaminated by the attention sink, with effective-rank distortions up to $721\times$ at 6.9B parameters. A single rank-1 SVD projection (*de-sinking*) removes the contamination and reveals monotonic rank *growth* rather than the widely reported “rank collapse.”

The Residual Stream Illusion: Why Standard Ablation Misleads in Vision-Language-Action Models

Sole author · [matched-random-ablation-E72D](#)

Zero ablation conflates a layer’s functional importance with the accumulated content of its residual stream, producing $25\times$ overestimates in VLAs. Introduces *matched-random* ablation (norm-matched random vectors) and *bypass* testing as forward-hook context managers; together they expose a token-role taxonomy and show that VLA fine-tuning structurally repurposes language tokens as routing nodes, with closed-loop validation on LIBERO.

Cross-View Attention Partition in Vision-Language-Action Models: A Quantitative Framework

Sole author · [sigma-u-3513](#)

Bypass VLAs (OpenVLA-OFT, π_0 , X-VLA) form strong asymmetric cross-view attention ($|\psi| \approx 0.33$), yet masking it costs only $\leq 17\%$ of the multi-view benefit — formation and function decouple. Traces this to a $\mathbb{Z}_2$ -symmetry violation in the Landau-framework training-data assumption and ships the ρ -probe as an architecture-invariant diagnostic, including a runtime monkey-patch of LlamaSdpaAttention for cross-view knockout.

Wrong-Object Long-Tail Failures in Vision-Language-Action Models

Co-first author · [vla-deploycheck-4982](#)

Average task-success cannot distinguish a harmless timeout from a confident wrong-object grasp. Builds a deployment audit (mutually-exclusive failure taxonomy, Wilson 95% CIs, two-stage cluster bootstrap, CI-style ship-gate) released as the `vla-deploycheck` CLI; demonstrates ship-blocking wrong-object failures on OpenVLA-OFT that look deployable on full-success metrics alone.

EXPERIENCE

Founder, Φ (fight) Research

May 2026 – present

- Independent research collective for physical-world AI fighting: adversarial multi-agent VLA, world models under contested dynamics, mechanistic interpretability of self-play policies.
- Three papers in progress for ICLR 2027 (co-authored with Han Muchen): one benchmark (Φ -Arena, co-led), one mechanistic interpretability paper (lead), one energy-bounded adversarial games paper (collaboration).

Founder, OfferI

Feb 2025 – Dec 2025 (*closed*)

- AI study-abroad school selection agent for international students.
- Built large-scale distributed crawler system and end-to-end agent workflow.

PROJECTS

Squirrel — AI Memory Layer

Dec 2025 – Feb 2026

Memory system for AI coding agents (Claude Code, Cursor, Codex). Rust daemon + MCP server; Python service for LLM-based memory extraction. Organization deleted at 1,000+ GitHub stars.

Xiaoniao — Cross-Platform Translation Tool (*open source*)

Aug – Sep 2025

Multi-platform translation tool in Go: Chrome extension + Windows app; three AI provider integrations.

EDUCATION

Hong Kong University of Science and Technology

Sep 2024 – 2029 (*expected*)

Electronic Engineering & Artificial Intelligence, Dual Major

– Top 3 in COMP6211 (PhD-level AI Startup Course), Summer 2025.

TECHNICAL SKILLS

Languages. Rust, Python, TypeScript, Go (proficient); comfortable picking up new languages quickly.

Research stack.

- *Mechanistic interpretability* for LLMs and VLAs — activation interventions (ablation, bypass, matched-random), attention analysis, linear probing, spectral analysis of hidden states.
- *Robotics evaluation* — closed-loop VLA rollouts in LIBERO / MuJoCo; deployment audits with failure-mode taxonomies and bootstrap / Wilson CIs.
- *Hands-on models* — OpenVLA(-OFT), π_0 , SmolVLA, X-VLA, ACT; GPT-2, Pythia, OLMo-2, Qwen2.5, LLaMA-2, LLaVA.
- *Stack* — PyTorch with HuggingFace Transformers / datasets / accelerate.

Hardware. LeKiwi mobile manipulator; dual-arm bimanual setups; embedded systems; mini-PC and SBC development.

Platforms. Native development across Windows, macOS, and Linux (Ubuntu, Kali, Fedora, NixOS, Arch).